P2 : Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis

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1. TITLE

Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis

Introduction

Urbanization has become one of the most significant drivers of environmental change across the world. Rapid growth of cities often leads to replacement of natural vegetation and open land with roads, buildings, concrete surfaces, and other impervious structures. These surfaces absorb and retain more solar energy than natural landscapes, causing urban regions to become significantly warmer than surrounding rural areas. This phenomenon is commonly known as the Urban Heat Island (UHI) effect.

In this study, Landsat-8 satellite imagery and Google Earth Engine were used to derive Land Surface Temperature and identify Urban Heat Island zones within Delhi. The relationship between built-up density and temperature distribution was examined using thermal and spectral analysis techniques. The generated outputs provide valuable insights into urban thermal hotspots and support sustainable urban planning and environmental management.

2. OBJECTIVE

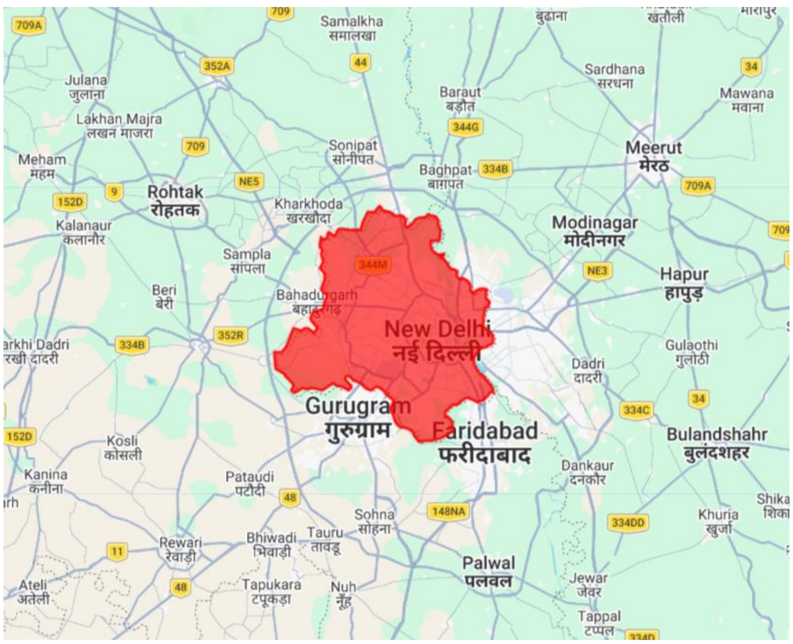
The primary objective of this project is to analyze the Urban Heat Island (UHI) effect in Delhi using satellite-derived Land Surface Temperature and built-up density indicators. The study aims to identify areas experiencing elevated surface temperatures and investigate their spatial relationship with urban development patterns.

The project demonstrates the application of Remote Sensing, GIS, and cloud-based geospatial analysis for addressing urban climate challenges and understanding the thermal characteristics of rapidly growing cities.

3. STUDY AREA

Study Area Description

The study area selected for this project is Delhi, the National Capital Territory (NCT) of India. Delhi is one of the largest metropolitan regions in the country and serves as a major administrative, economic, educational, and transportation hub. Rapid urban expansion, increasing population density, and extensive infrastructure development have significantly transformed the natural landscape of the city over the past few decades.



The combination of rapid urbanization, population growth, and environmental stress makes Delhi a suitable case study for Urban Heat Island assessment using satellite remote sensing techniques.

Location Details

Parameter	***Details***
Study Area	*Delhi*
Country	*India*
Latitude	*Approximately 28.61° N*
Longitude	*Approximately 77.20° E*
Climate Type	*Semi-Arid / Subtropical*
Analysis Type	*Urban Heat Island Assessment*

Area of Interest (AOI):

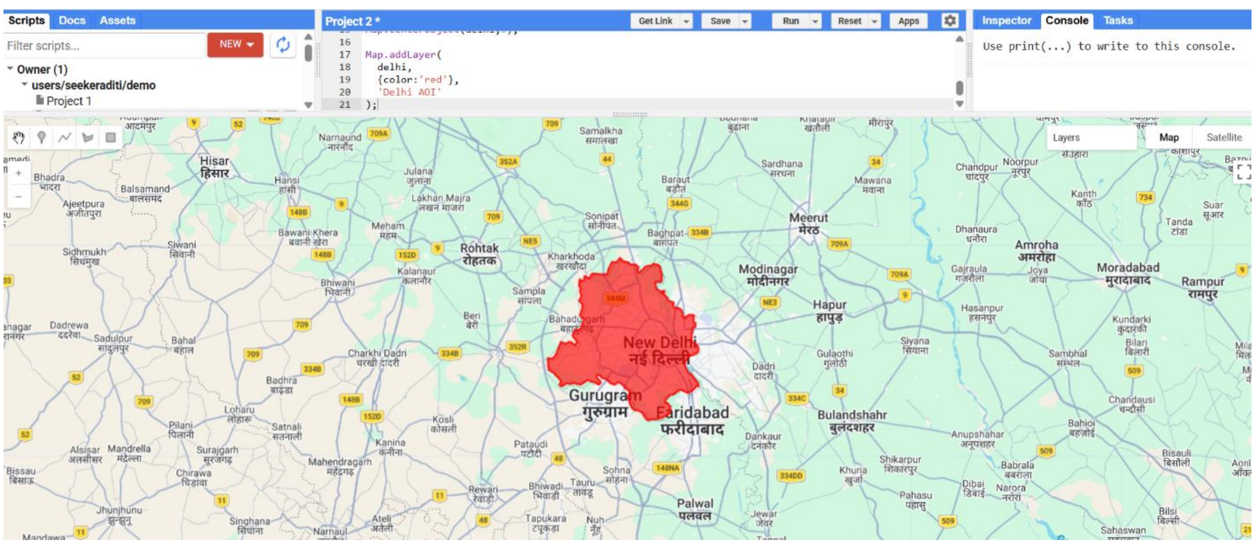


__Figure 1:__ Area of Interest (AOI) showing the administrative boundary of Delhi used for Urban Heat Island analysis.

4. DATA USED

Satellite Data

The study utilized Landsat-8 Collection 2 Level-2 imagery available through Google Earth Engine. Landsat-8 provides multispectral and thermal observations suitable for vegetation analysis, built-up density mapping, and Land Surface Temperature estimation.

The imagery was selected for the year 2025 with cloud cover less than 10% to ensure reliable thermal observations and minimize atmospheric interference.

Satellite Specifications

Parameter	*Details*
Satellite	*Landsat-8*
Sensor	*OLI/TIRS*
Dataset	*Collection 2 Level-2*
Spatial Resolution	*30 m*
Analysis Year	*2025*
Cloud Cover	*Less than 10%*

Bands Used

Band	*Purpose*
SR_B4	*Red Band*
SR_B5	*Near Infrared (NIR)*
SR_B6	*Shortwave Infrared (SWIR)*

Band	*Purpose*
ST_B10	*Thermal Infrared Band*

Software Used

Google Earth Engine (GEE)

QGIS

Google Earth Pro (for visual reference)

Data Sources

Landsat Data:
https://earthexplorer.usgs.gov/

Google Earth Engine Data Catalog:
https://developers.google.com/earth-engine/datasets

Administrative Boundary Data:
https://gadm.org/

Reference Imagery:
https://www.google.com/earth/versions/#earth-pro

5. METHODOLOGY

Overview of Methodology

The Urban Heat Island (UHI) assessment was carried out using Landsat-8 satellite imagery and cloud-based geospatial processing within Google Earth Engine (GEE). The workflow involved acquisition of satellite data, generation of vegetation and built-up indices, estimation of Land Surface Temperature (LST), identification of Urban Heat Island zones, and computation of area statistics.

The methodology was designed to evaluate the relationship between urban development and surface temperature distribution across Delhi.

The overall workflow followed is shown below:

Satellite Data Acquisition → AOI Selection → NDVI Calculation → NDBI Calculation → LST Estimation → UHI Identification → Area Statistics → Result Interpretation

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Step 1: Area of Interest (AOI) Selection

The first step involved defining the Area of Interest (AOI) representing Delhi. Administrative boundary data available within Google Earth Engine were used to extract the study region.

The AOI serves as a spatial mask for all subsequent analyses and ensures that calculations are restricted only to the selected study area.

Purpose

Define study boundary

Limit analysis to Delhi

Improve computational efficiency

Ensure accurate statistics generation

Output

Administrative boundary of Delhi displayed on the map.

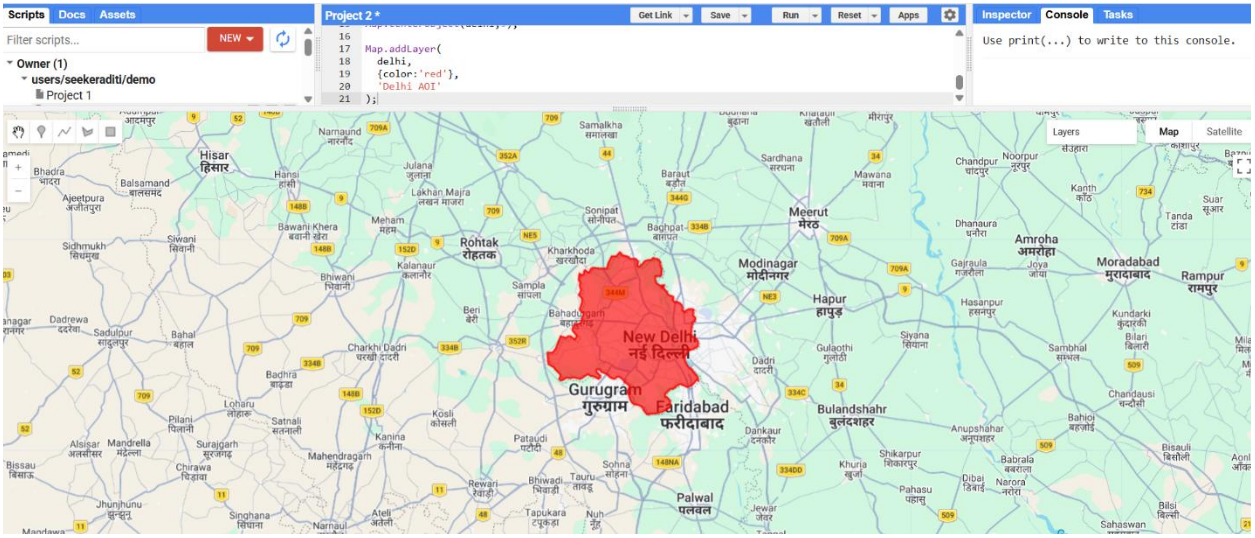

Figure 2: *Administrative boundary of Delhi used as the Area of Interest (AOI).*

Step 2: Landsat-8 Data Acquisition

Cloud-free Landsat-8 Collection 2 Level-2 imagery was acquired from the Google Earth Engine Data Catalog.

Images from the year 2025 were filtered based on:

Study area boundary

Date range

Cloud cover less than 10%

A median composite image was generated to reduce atmospheric noise and improve image quality.

Dataset Information

Parameter	***Value***
Satellite	*Landsat-8*

Parameter	***Value***
Collection	*Collection 2 Level-2*
Year	*2025*
Cloud Cover	*<10%*
Resolution	*30 m*

Purpose

Obtain multispectral and thermal imagery

Ensure cloud-free observations

Create representative annual composite

Output

Processed Landsat-8 image collection.

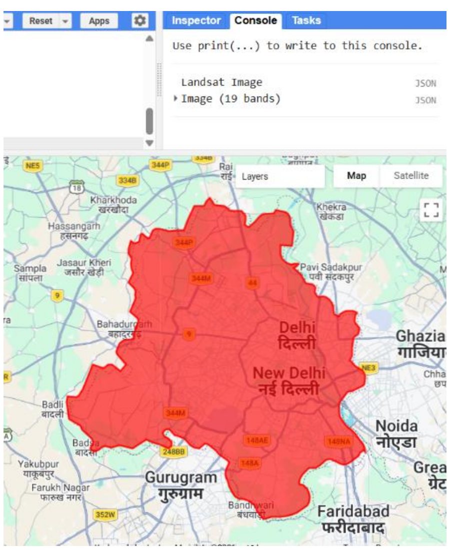

Figure 3: *Landsat-8 imagery used for thermal and spectral analysis.*

Step 3: NDVI Calculation

The Normalized Difference Vegetation Index (NDVI) was calculated to identify vegetation distribution within the study area.

NDVI is one of the most widely used vegetation indices in remote sensing and helps distinguish vegetated areas from built-up and barren surfaces.

Higher NDVI values indicate healthy vegetation, whereas lower values indicate urban surfaces, bare land, or water bodies.

Formula used:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Where:

NIR = Near Infrared Band (Band 5)

Red = Red Band (Band 4)

Importance of NDVI:

Identifies vegetation cover

Supports interpretation of thermal patterns

Helps understand cooling effects of green spaces

Interpretation

NDVI Range	***Interpretation***
< 0	*Water/Non-vegetation*
0 – 0.2	*Sparse vegetation*
0.2 – 0.5	*Moderate vegetation*

NDVI Range	Interpretation
> 0.5	Dense vegetation

Output

NDVI map showing vegetation distribution across Delhi.

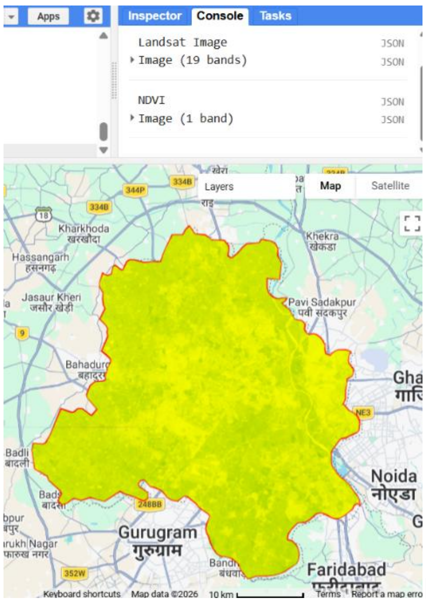

Figure 4: NDVI map generated from Landsat-8 imagery.

Step 4: Built-up Density Analysis using NDBI

To assess urban development intensity, the Normalized Difference Built-up Index (NDBI) was calculated.

NDBI is commonly used to identify built-up surfaces such as roads, residential areas, commercial zones, and industrial infrastructure.

Higher NDBI values generally represent urbanized areas, while lower values correspond to vegetation and open land.

Formula used:

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$$

Where:

SWIR = Shortwave Infrared Band (Band 6)

NIR = Near Infrared Band (Band 5)

Importance of NDBI:

Identifies built-up regions

Measures urban expansion

Supports Urban Heat Island analysis

Interpretation

NDBI Value	***Interpretation***
Negative	*Vegetation or water*
Near Zero	*Mixed land cover*
Positive	*Built-up surfaces*

Output:

Built-up density distribution map.

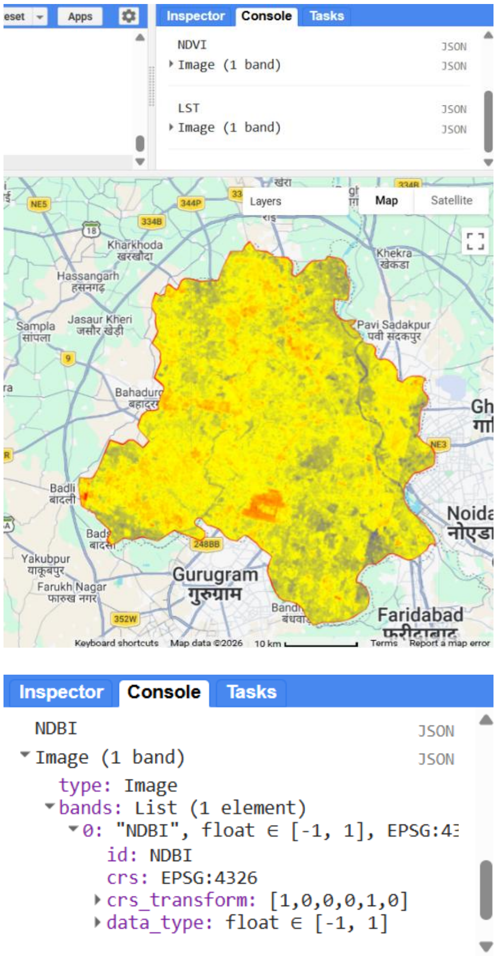

Figure 5: *NDBI-based built-up density map of Delhi.*

Step 5: Land Surface Temperature (LST) Estimation

Land Surface Temperature (LST) was derived from the thermal infrared band (ST_B10) available within the Landsat-8 Level-2 product.

The thermal band values were converted into temperature values and expressed in degrees Celsius.

LST provides information regarding the thermal characteristics of the Earth's surface and serves as the primary indicator for Urban Heat Island analysis.

Importance of LST

Measures surface temperature

Detects thermal hotspots

Supports climate and urban studies

Temperature Classification

Temperature Category	***Description***
Low	*Cooler surfaces*
Moderate	*Transitional zones*
High	*Urban Heat Island regions*

Output

Spatial distribution of land surface temperature across Delhi.

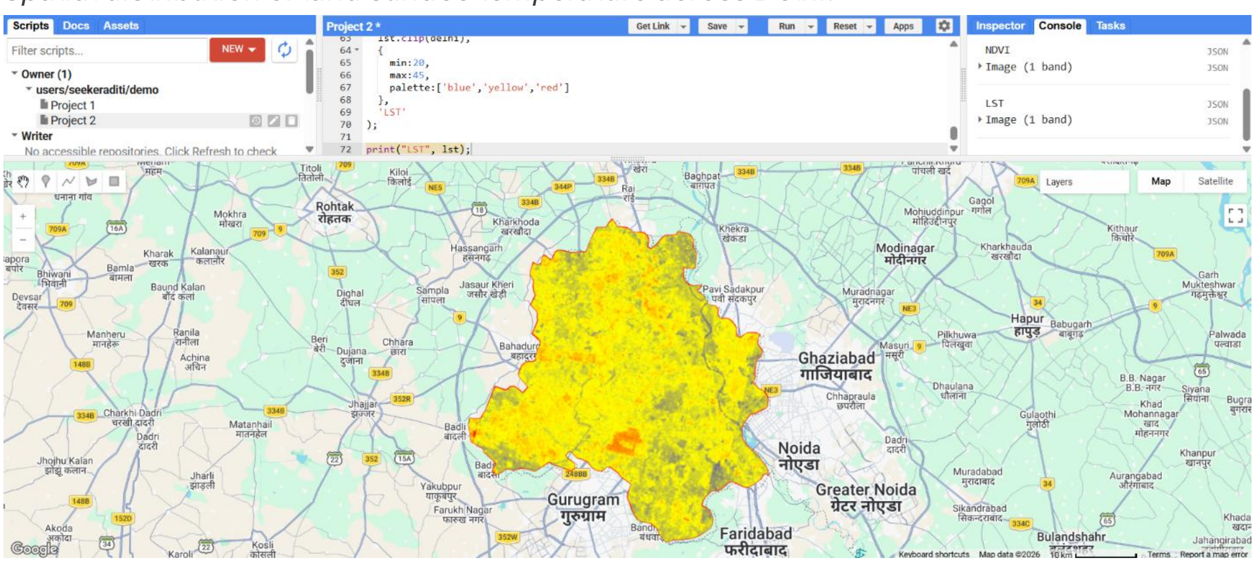

Figure 6: *Land Surface Temperature (LST) distribution derived from Landsat-8 thermal imagery.*

Step 6: Urban Heat Island Identification

Urban Heat Island (UHI) zones were identified by applying a temperature threshold to the Land Surface Temperature layer.

Areas exhibiting elevated temperatures were classified as Urban Heat Island zones.

These hotspots generally correspond to densely built-up regions, transportation infrastructure, industrial zones, and areas with limited vegetation cover.

Purpose

Identify high-temperature regions

Delineate urban thermal hotspots

Assess thermal stress zones

Output

Urban Heat Island map highlighting temperature hotspots.

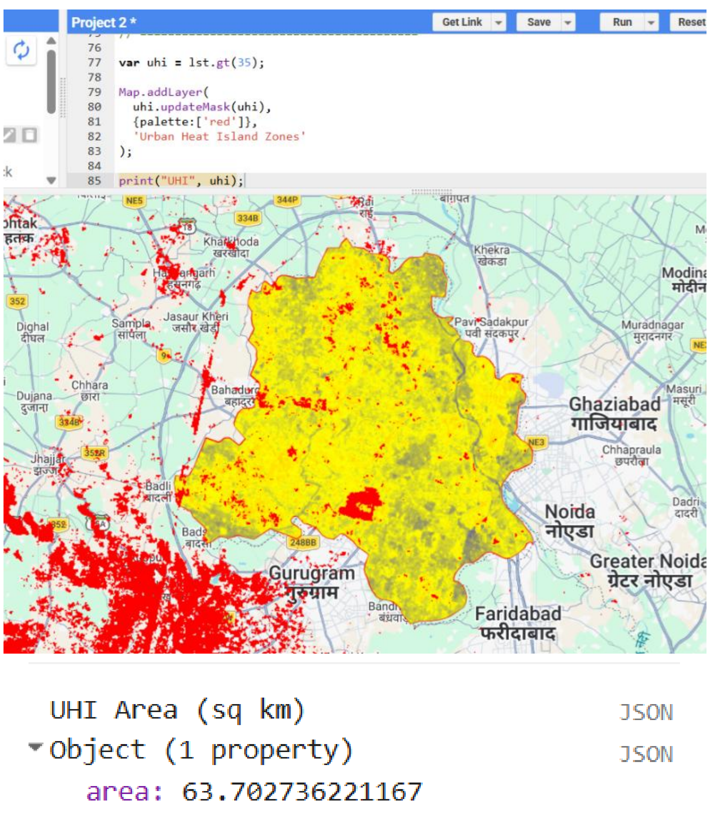

Figure 7: *Urban Heat Island zones identified from Land Surface Temperature analysis.*

Step 7: Area Statistics

The final step involved calculating the spatial extent of Urban Heat Island zones.

Area statistics were generated using pixel-based calculations within Google Earth Engine. The total area occupied by identified UHI regions was computed and expressed in square kilometers.

Calculated Result

Parameter	***Value***
Urban Heat Island Area	*63.70 sq km*

Importance

Quantifies thermal hotspot extent

Supports comparison with future studies

Provides measurable urban climate indicators

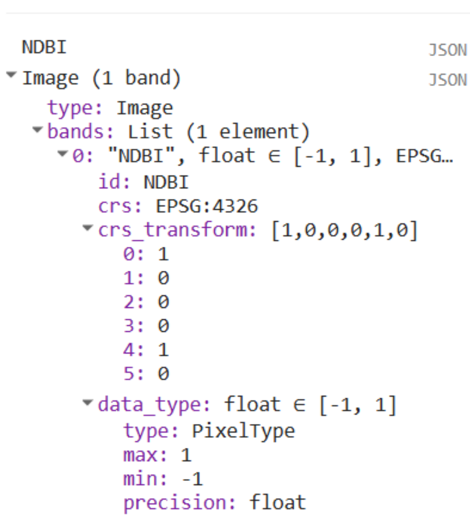

Figure 8: *Area statistics generated for Urban Heat Island zones.*

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6. RESULTS AND DISCUSSION

6.1 NDVI Analysis

The Normalized Difference Vegetation Index (NDVI) was generated using Landsat-8 Red and Near Infrared bands to assess vegetation distribution within Delhi. The resulting NDVI map showed considerable variation in vegetation density across the study area.

Higher NDVI values were primarily observed in parks, forest patches, agricultural fields, and river-adjacent green spaces. Lower NDVI values were associated with densely urbanized regions, transportation infrastructure, industrial areas, and impervious surfaces.

The analysis demonstrated that vegetation cover is unevenly distributed across Delhi. Areas with dense vegetation generally exhibited lower surface temperatures, highlighting the cooling effect of green spaces on the urban environment.

Observation

High NDVI values indicate healthy vegetation.

Low NDVI values correspond to built-up and barren areas.

Vegetated zones contribute to reduction in surface temperature.

NDVI serves as an important indicator for understanding thermal variations.

6.2 Built-up Density Analysis using NDBI

The Normalized Difference Built-up Index (NDBI) was generated to identify urbanized and impervious surfaces within the study area.

Positive NDBI values were predominantly observed in densely developed urban regions, commercial centers, transportation corridors, and industrial areas. Negative values represented vegetation-covered and open land regions.

The NDBI map clearly highlighted the spatial distribution of built-up density across Delhi and provided valuable insights into urban expansion patterns.

Observation

Higher NDBI values indicate urbanized surfaces.

Lower values represent vegetation and open spaces.

Built-up areas are concentrated in central and highly developed regions.

Urban development patterns strongly influence local temperature conditions.

6.3 Land Surface Temperature (LST) Analysis

Land Surface Temperature was derived from Landsat-8 thermal infrared data using the ST_B10 thermal band. The generated LST map revealed substantial temperature variation across the study area.

Higher temperatures were observed in densely urbanized and built-up regions, while relatively lower temperatures were associated with vegetated areas and open spaces.

The temperature distribution pattern indicated the presence of localized thermal hotspots resulting from urban development and reduced vegetation cover.

Observation

High temperatures were concentrated in urbanized regions.

Cooler temperatures occurred near vegetated areas.

Thermal variation reflects differences in land cover characteristics.

LST effectively highlights urban climate conditions.

6.4 Urban Heat Island (UHI) Zone Identification

Urban Heat Island zones were identified using the Land Surface Temperature layer by applying a temperature threshold to isolate thermal hotspot regions.

The resulting UHI map showed clusters of elevated temperature concentrated in highly urbanized parts of Delhi. These hotspots generally corresponded to dense built-up regions *characterized by extensive concrete surfaces, transportation infrastructure, and reduced vegetation.*

Urban Heat Island zones were visually distinct from surrounding cooler regions and clearly demonstrated the impact of urbanization on thermal conditions.

Observation

Urban Heat Islands are concentrated in highly developed areas.

Vegetation-rich zones exhibit lower thermal intensity.

Urban infrastructure contributes significantly to heat accumulation.

Spatial distribution of UHI zones reflects land-use patterns.

6.5 Area Statistics

Area statistics were computed to quantify the spatial extent of Urban Heat Island zones within Delhi.

The analysis indicated that approximately ***63.70 square kilometers*** *of the study area falls under Urban Heat Island conditions.*

Area Statistics Table

Parameter	***Value***
Urban Heat Island Area	*63.70 sq km*

```
NDBI                                              JSON
▾ Image (1 band)                                  JSON
    type: Image
  ▾ bands: List (1 element)
    ▾ 0: "NDBI", float ∈ [-1, 1], EPSG…
        id: NDBI
        crs: EPSG:4326
      ▾ crs_transform: [1,0,0,0,1,0]
          0: 1
          1: 0
          2: 0
          3: 0
          4: 1
          5: 0
      ▾ data_type: float ∈ [-1, 1]
          type: PixelType
          max: 1
          min: -1
          precision: float
```


Figure 13: *Area statistics generated using Google Earth Engine.*

Interpretation

The calculated area indicates a significant extent of thermally stressed urban regions within Delhi. Such hotspots may contribute to increased energy demand, heat-related health risks, and environmental stress during summer periods. The findings highlight the need for sustainable urban planning measures such as increasing vegetation cover, promoting green infrastructure, and preserving urban open spaces.

6.6 Relationship Between Built-up Density and Temperature

A comparative assessment of NDBI and LST layers revealed a strong spatial relationship between built-up density and surface temperature.

Areas exhibiting higher NDBI values generally corresponded with elevated Land Surface Temperature values. Conversely, regions with lower built-up density and greater vegetation cover tended to exhibit lower temperatures.

This relationship demonstrates that urbanization plays a major role in modifying local thermal environments. Impervious surfaces absorb and store solar radiation during the day and release heat slowly, resulting in increased surface temperatures and Urban Heat Island formation.

Key Findings

Built-up areas showed higher thermal intensity.

Vegetated areas displayed lower temperatures.

Urban Heat Islands were closely associated with dense infrastructure.

NDBI and LST exhibited a positive spatial relationship.

Urban planning strategies can reduce thermal stress through increased vegetation cover.

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7. CONCLUSION

This project successfully demonstrated the application of Remote Sensing, Geographic Information Systems (GIS), and Google Earth Engine for Urban Heat Island assessment in Delhi. Landsat-8 satellite imagery was utilized to derive Land Surface Temperature (LST), vegetation information through NDVI, and built-up density using NDBI.

The results revealed significant spatial variation in surface temperature across the study area. Urban Heat Island zones were primarily concentrated within densely built-up regions characterized by extensive impervious surfaces and limited vegetation cover. The analysis further indicated a clear relationship between built-up density and elevated surface temperatures.

*The generated NDVI map highlighted the cooling influence of vegetation, while the NDBI analysis effectively identified urbanized regions contributing to thermal hotspot formation. The calculated Urban Heat Island area of approximately **63.70 square kilometers** reflects the extent of thermally stressed urban zones within Delhi.*

The study demonstrates the effectiveness of satellite-based thermal analysis for urban environmental monitoring and climate-related assessments. Such information can support *urban planners, environmental managers, and policymakers in developing strategies to reduce heat stress, improve urban livability, and enhance climate resilience.*

Although the analysis relied on a single-year Landsat composite and simplified threshold-based classification, the overall workflow provides a reliable framework for Urban Heat Island mapping. Future studies may incorporate multi-year datasets, seasonal comparisons, advanced thermal modeling techniques, and machine learning approaches to improve accuracy and support long-term urban climate monitoring.

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8. REFERENCES

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Google Earth Engine Data Catalog.
https://developers.google.com/earth-engine/datasets

GADM Administrative Boundary Database.
https://gadm.org/

Google Earth Pro.
https://www.google.com/earth/versions/#earth-pro

NASA Earth Observatory. Urban Heat Island Resources.
https://earthobservatory.nasa.gov/

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Weng, Q. (2009). Thermal Infrared Remote Sensing for Urban Climate and Environmental Studies.

Voogt, J. A., & Oke, T. R. (2003). Thermal Remote Sensing of Urban Climates.